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METHOD FOR MAKING A PLANT-BASED ZERO SUGAR FOOD PRODUCT

Abstract

A method for making a plant-based food product with zero sugar is provided. The plant-based food product includes an amount of water, a sugar content of between 0 wt % and 0.4 wt % referred to as having “zero sugar,” and a plant-based fiber product. The plant-based fiber product includes an amount of water, a plant-based flour, an amount of plant fiber, and a plurality of enzymes comprising: an alpha amylase, a glucose oxidase, catalase, and a cellulase. The plant-based food product may be incorporated in a milk, yogurt, shake, or bar, for example.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is related to and claims priority to each of the following: U.S. Provisional Patent Application Ser. No. 63/134,047, filed on Jan. 5, 2021; U.S. Nonprovisional patent application Ser. No. 17/194,947, filed Mar. 8, 2021 (now U.S. Pat. No. 11,771,121 issued on Oct. 3, 2023); and U.S. Nonprovisional patent application Ser. No. 18/480,294, filed Oct. 3, 2023, the entire contents of each being herein incorporated by reference.

TECHNICAL FIELD

[0002] This application relates to food products and methods and, in particular, relates to plant-based food products having zero sugar content and methods of manufacturing plant-based food products having zero sugar content.

BACKGROUND

[0003] In the manufacture of food products, reducing the sugar content of food products has been an increasingly appealing field of development based on the consumer marketplace and its demands. Consumer food choice decisions often turn on the sugar content in food for a variety of reasons. Popular diets, diabetic consumers, and/or other reasons have driven consumers to search for food products having lower sugar content than their predecessors. Consumption of high-sugar food products has been shown to increase a consumer's risk of developing diabetes, weight gain, and other negative health consequences. It is therefore desirable for food product manufacturers to offer alternatives and further developments for food products so as to provide a lower sugar content as an option.

[0004] Plant-based food products have historically needed to add sugar/carbohydrates to assist in improving flavoring, color, and desirability of consuming these plant-based food products. While the consumption of plant-based food products provides a significant source of many nutrients found in the plant bases of plant-based food products, as mentioned above, the addition of sugar can be detrimental to the health of the consumer and many consumers are choosing food options with reduced or zero sugar. As a result, many consumers that would otherwise be interested in plant-based food products are left with no good options in the current food product marketplace based on typical levels of sugar content in such conventional products.

[0005] Accordingly, it would be desirable to produce plant-based food products having reduced or little-to-no sugar in the final product. Of course, maintaining previously-made improvements in flavoring, color, and desirability of such food products is also an objective when making alternatives with lower sugar content.

SUMMARY

[0006] To address these and other technical problems within the conventional art, a plant-based food product with zero sugar is provided in accordance with one embodiment. The food product includes a first amount of water, a sugar content of between 0 wt % and 0.4 wt %, and a plant-based fiber product. The plant-based fiber product includes a second amount of water, a plant-based flour, an amount of plant fiber, and a plurality of enzymes. These enzymes may include at least an alpha amylase, a glucose oxidase, catalase, and a cellulase. This combination of elements making up the plant-based food product allows for various types of food products including plant-based

milks, yogurts, shakes, bars, and the like with desirable flavor and coloring at least similar to conventional products with high sugar content, while substantially eliminating all sugar content from the food product. As such, the plant-based food product of this and other embodiments improves the field by providing more consumer options for such types of food products without the usual downside of a high sugar content.

[0007] In the context of this application, it will be understood that the label “zero sugar” is applied so as to mean zero or substantially zero sugar content in the final food product generated, which for the sake of example, may be defined as a sugar content of between 0 wt % and 0.4 wt %.

[0008] A plant-based food product is provided. The plant-based food product includes a first amount of water, a sugar content of between 0 wt % and 0.4 wt %, and a plant-based fiber product. The plant-based fiber product includes a second amount of water, a plant-based flour, an amount of plant fiber, and a plurality of enzymes including an alpha amylase, a glucose oxidase, catalase, and a cellulase.

[0009] In another embodiment according to the invention, the plant-based flour is oat flour.

[0010] In another embodiment according to the invention, the plant-based food product further includes an added flavoring.

[0011] In another embodiment according to the invention, the plant-based food product further includes a second plant-based flour selected from a group consisting of buckwheat flour, brown rice flour, potato flour, barley flour, oat flour, pumpkin flour, chia flour, yellow pea flour, and mixtures thereof.

[0012] In another embodiment according to the invention, the plant-based food product further includes an additional enzyme comprising a protease

[0013] In another embodiment according to the invention, the plant-based food product further includes an additional enzyme including an acyl transferase made from *Streptomyces mobaraensis* bacteria using microbial fermentation.

[0014] In another embodiment according to the invention, the plant-based food product further includes maltodextrin made from starch derived from tapioca.

[0015] In another embodiment according to the invention, the plant-based food product further includes an isolate selected from soy, pea, rice, pumpkin, potato, buckwheat, chia, amaranth, brown rice, fava bean, barley, oat, mushrooms, fungi, and mixtures thereof.

[0016] In another embodiment according to the invention, the plant-based food product further includes a rice protein concentrate.

[0017] In another embodiment according to the invention, the plant-based food product further includes an oil selected from a group consisting of medium chain triglyceride oil, canola oil, and mixtures thereof.

[0018] In another embodiment according to the invention, the plant-based food product further includes gellan gum.

[0019] In another embodiment according to the invention, the plant-based food product further includes a vitamin selected from a group consisting of vitamin A, vitamin D, Vitamin E, Vitamin K, Vitamin C and Vitamins B1, B2, B3, B5, B6, B7, B9, B12, and mixtures thereof.

[0020] In another embodiment according to the invention, the plant-based food product includes a plant fiber being oat fiber.

[0021] A method of making a plant-based food product is also provided. The method includes forming a plant-based fiber product and adding water to the plant-based fiber product to form the plant-based food product. The plant-based food product formed has a sugar content of between 0 wt % and 0.4 wt %. In the described method, the plant-based fiber product is formed by adding a plant-based flour to water to form a wetted plant-flour mixture, adding an alpha amylase to the wetted plant-flour mixture, heating the wetted plant-flour mixture to a temperature between 180° F. and 200° F., cooling the wetted plant-flour mixture to a temperature between 130° F. and 150° F., adding a glucose oxidase to the wetted plant-flour mixture, adding a catalase to the wetted plant-

flour mixture, clarifying the wetted plant-flour mixture into a plant fiber product and a base product, decantering the plant fiber product to form a decantered plant fiber product, mixing the decantered plant fiber product with the base product to form a buffered product, mixing the decantered plant fiber product with the base product to form a buffered product, sanitizing the buffered product by steam injection sanitation to form a sanitized buffered product, and maintaining the sanitized buffered product for a time between 3 hours and 5 hours to form the plant-based fiber product.

[0022] In another embodiment according to the invention, the plant-based flour used in the method is oat flour.

[0023] In another embodiment according to the invention, the method further includes adding an additional enzyme including a protease to the plant-based food product.

[0024] In another embodiment according to the invention, the method further includes adding an additional enzyme including an acyl transferase made from *Streptomyces mobaraensis* bacteria using microbial fermentation to the plant-based food product

[0025] In another embodiment according to the invention, the method further includes adding maltodextrin made from starch derived from tapioca to the plant-based food product.

[0026] In another embodiment according to the invention, sanitizing the plant-based fiber product is done by a steam injection sanitation process to form a sanitized plant-based fiber product. The steam injection sanitation process includes feeding the buffered product to a direct steam injection feed tank, preheating the buffered product to a temperature of between 170° F. and 190° F., exposing the heated buffered product to direct steam injection, further raising a temperature of the buffered product to between 280° F. and 290° F., and maintained at the temperature for at least 4 seconds, and cooling the buffered product to between 40° F. and 50° F. to produce the sanitized plant-based fiber product.

[0027] In another embodiment according to the invention, the method includes fruit to the plant-based food product.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] Further features, advantages and possible applications of the present invention will be apparent from the following description in connection with the figures, in which the same reference signs are used throughout for the same or mutually corresponding elements of the invention. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more embodiments of the invention and, together with the general description given above and the detailed description given below, explain the one or more embodiments of the invention.

[0029] FIG. 1 is a flow chart that illustrates a first sub-method and a second sub-method of a method of manufacturing a plant-based food product.

[0030] FIG. 2 is a process diagram that shows a method of manufacturing a plant-based food product.

DETAILED DESCRIPTION

[0031] According to embodiments of the invention summarized herein, the method includes a first sub-method and a second sub-method that cooperate to form a plant-based fiber product. The plant-based fiber product is combined with water, and optionally, with additional components, to form the plant-based food product. The method described herein removes essentially all sugars from the plant-based materials resulting in a plant-based food product having zero sugar content (“zero sugar” being as defined herein). All ranges of parameters described herein include the endpoints of the ranges.

[0032] As used herein, the terms “zero sugar”, “sugar free”, “essentially all sugars” indicates that the product complies with FDA Regulations stating that “zero sugar” means the food must be less than 0.5 g sugar per labelled serving (i.e., 5.3 oz/150 g) and per 6 oz/170 g (the RACC or serving size for yogurt). Accordingly, food products described herein as “zero sugar”, “sugar free”, or free of “essentially all sugar”, have, at most 0.5 g sugar per 5.3 oz or 150 g of food product or, in the case of a yogurt, 0.5 g sugar per labelled serving of 6 oz or 170 g of yogurt. In addition, food products described herein as “zero sugar”, “sugar free”, or free of “essentially all sugar”, include a sugar content in an amount of between 0 wt % and 0.4 wt % of the food product.

[0033] Referring to FIGS. 1 and 2, a method **100** of manufacturing a plant-based food product is provided and a system **200** for manufacturing the plant-based food product is also provided. The method **100** includes a first sub-method and a second sub-method. The first sub-method forms a plant-based fiber product. The first sub-method includes a step **102** of providing water in an ND base tank **206**. The first sub-method may optionally include step **104**, step **108**, or both steps **104**, **108** of adding enzymes to the water provided in step **102**. Furthermore, in step **106**, plant-based flour is added to the water from step **102** to form a wetted plant-flour mixture. In some examples, such as the example shown in FIG. 1, the enzymes may be added stepwise. That is, a first portion of the enzymes may be added to the water, for example in step **104**, before introducing the plant-based flour in step **106**, and a second portion may be added to the mixture, for example in step **108**, after the plant-based flour is introduced. The first portion may be, for example, about 50% of the enzymes, and the second portion may be, for example, about 50% of the enzymes, or the balance of enzymes that remains after the addition of the first portion. The enzyme added at steps **104**, **108** is CSE-10, which is an alpha amylase. The CSE-10 specifically digests starch to make 3 unit carbohydrates. Specifically, the CSE-10 digests starch naturally present in oats. This naturally present starch is digested into maltotriose and other high dextrose equivalent starches. In one example, the alpha amylase used in CSE-10 is manufactured by Koninklijke DSM N.V. (Royal DSM, referred to herein as DSM) located in Heerlen, Netherlands under the product name Validase BAA 1000L.

[0034] In step **106**, plant-based flour is added to the water from step **102** which may further include the enzymes added in step **104**. As described above, the plant-based flour may be oat flour. In some examples, the plant-based flour may be selected from a group consisting of oat flour, corn flour, pea flour, soy flour, buckwheat flour, wheat flour, potato flour, tapioca flour, and rice flour. The amount of plant-based flour added is between 22 and 35 wt %, preferably, the amount of plant-based flour added results in about 28 wt % of the resulting slurry to be plant-based flour.

[0035] The first sub-method further includes, in step **110**, heating the wetted plant-flour mixture. In some embodiments, the heating may result in the plant-flour mixture to be heated, in step **110**, to between 180° F. and 200° F., preferably to about 194° F.

[0036] The first sub-method further includes recirculating the wetted plant-flour mixture, in step **112**, with or without enzymes added, for 15 minutes to 25 minutes, preferably 20 minutes, maintaining the heated temperature of the wetted plant-flour mixture throughout the recirculating process. The temperature of the wetted plant-flour mixture is maintained via the heat exchanger **208** and is mixed via liquefier **202**. The recirculation is propelled by pump **204** and is fed back into the ND base tank **206**.

[0037] The first sub-method further includes cooling the wetted plant-flour mixture, in step **114**, to a temperature between 130° F. and 150° F., preferably 140° F.

[0038] Optionally, such as in the example method **100** shown in FIG. 1, the first sub-method may include adding additional enzymes to the cooled wetted plant-flour mixture, in step **116**. These additional enzymes may be the same or different than the enzymes optionally added previously to the wetted plant-flour mixture in steps **104**, **108**. In some embodiments, the enzymes added at this step may be CSE-6 and CSE-4. CSE-6 is a sugar enzyme including glucose oxidase, which in one example is manufactured by DSM under the product name Delvolase GO 10. CSE-4 is a sugar

enzyme including catalase, which in one example is manufactured by DSM under the product name Maxibright. In some examples, CSE-10 is a maltotriose that specifically breaks down starch included in the plant-based flour into repeating sub units having a degree of polymerization of 3 or higher. CSE-6 is an enzyme that converts any glucose produced in the above-described process into a glucuronic acid or a salt of glucuronic acid. CSE-4 is an enzyme that converts hydrogen peroxide present, such as hydrogen peroxide produced by the action of CSE-6, into oxygen. The oxygen produced by this method helps to enhance the functionality of the CSE-6 enzyme.

[0039] The first sub-method further includes further cooling the wetted plant-flour mixture to 110° F. to 130° F., preferably 120° F., in step **118**. The wetted plant-flour mixture is cooled in feed tank **210**.

[0040] The first sub-method further includes clarifying the wetted plant-flour mixture, in step **120**, in a clarifier **212**. The clarifier separates the wetted plant-flour mixture into a stream of plant fiber product and a stream of base product. As will be described below in the Examples, the base product may include water between 69 wt % and 74 wt %, plant-based flour between 26 wt % and 31 wt %, CSE-10 between 0.015 wt % and 0.0350 wt %, CSE-6 between 0.0015 wt % and 0.0035 wt %, and CSE-4 between 0.0015 wt % and 0.004 wt %. As will also be described below in the Examples, the plant fiber product includes a fiber sludge constituting between 90 wt % and 99 wt % of the plant fiber product. The fiber sludge includes between 25 wt % and 28 wt % solids after the clarifying step **120** in the clarifier **212**. The plant fiber product further includes CFE-1 in an amount between 0.1 wt % and 0.25 wt %. CFE-1 is a fiber enzyme including cellulase, which in one example is manufactured by DSM under the product name Validase TRL. The plant fiber product further includes CSE-6 in an amount between 0.005 wt % and 0.01 wt %, and CSE-4 in an amount between 0.0005 wt % and 0.01 wt % of the plant fiber product. The plant fiber product further includes water in an amount between 0.73 wt % and 9.89 wt % of the plant fiber product.

[0041] The first sub-method may further include filtering the plant fiber product through a decanter **213** to form a decanted plant fiber product stream in step **122**. Optionally, the first sub-method further includes adding enzymes to a tank in steps **130**, **132**, **134**. The enzymes added at steps **130**, **132**, **134** may include CFE-1, CSE-6, and CSE-4. CFE-1 is a fiber enzyme including cellulase, which in one example is manufactured by DSM under the product name Validase TRL. As explained above, CSE-6 is a sugar enzyme including glucose oxidase, such as manufactured by DSM under the product name Delvolase GO 10. CSE-4 is a sugar enzyme including catalase, such as manufactured by DSM under the product name Maxibright. Optionally, in step **126**, fiber solids included in the plant-based flour are treated with CFE-1, CSE-4, and CSE-6 enzymes. The addition of these enzymes enables the use of the fiber solids to be incorporated into the buffer tank **214**. Step **126** occurs after step **122**, and may occur in a separate tank, where the clarified and decanted insoluble plant fiber and water are mixed to form a 20-25% slurry. The slurry is then treated with CFE-1, CSE-4, and CSE-6 to form a soluble fiber solution that is then injected back into the process before step **128** to make a plant-based slurry, which may be whole oat slurry, and is then fed through the DSI process, in step **124**. The plant fiber product and the base product are mixed and fed into a buffer tank **214**, thus beginning the second sub-method.

[0042] The second sub-method forms a plant-based fiber product. The second sub-method includes a direct steam injection (DSI) sanitation process. The DSI sanitation process includes combining at least a portion of the decanted plant fiber product with at least a portion of the base product to form a base product mixture in, for example, a buffer tank **214**, to form a buffered product. The buffered product is fed to a DSI feed tank **216**. The buffered product is then preheated to a temperature of between 170° F. and 190° F., preferably 180° F., in preheater **218**. The thus heated buffer product is then exposed to direct steam injection, further raising the temperature of the buffered product to between 280° F. and 290° F., preferably 282° F., in the steam heater **220**. The buffered product is maintained at the steam-heated temperature for at least 4 seconds. The heated buffered product may be allowed to cool, and then reheated to between 280° F. and 290° F.,

preferably 282° F., in steam heater **220**. This heating/cooling cycle may be repeated as necessary to form a sanitized buffered product. The sanitized buffered product is cooled to between 40° F. and 50° F., preferably 45° F., in cooler **222**, and fed into a tank **224**, and accordingly ending the DSI sanitation process in step **124**.

[0043] The tank **224** mixture is permitted to stand in step **136**, optionally with maintaining the temperature of the tank mixture, for a time between 3 hours and 5 hours, preferably 4 hours, and forms the plant-based fiber product, in step **138**. In the tank **224**, the sanitized buffered product is heated to between 110° F. and 130° F., preferably 122° F., in step **128**.

[0044] The plant-based fiber product is formed into the plant-based food product by adding at least water to the plant-based fiber product. Optionally, further additions may be added to the plant-based fiber product, such as flavorings (artificial or natural), plant protein isolates and/or concentrates such as from soy, pea, rice, pumpkin, potato, buckwheat, chia, amaranth, brown rice, fava bean, barley, oat, mushrooms, fungi, and mixtures thereof, enzymes, plant-based flour such as buckwheat flour, brown rice flour, potato flour, barley flour, oat flour, pumpkin flour, chia flour, yellow pea flour, and mixtures thereof, oils such as medium chain triglyceride (MCT) oil which include two or three fatty acids, canola oil, and mixtures thereof, gellan gum, and vitamins such as Vitamin A, Vitamin D, Vitamin E, Vitamin K, Vitamin C and Vitamins B1, B2, B3, B5, B6, B7, B9, B12, and mixtures thereof. The fatty acids included in the MCT oil include carbon chain moieties having carbon chains of between 6 and 13 carbon atoms, and may include branched or non-branched carbon chains, saturated or unsaturated carbon chains, or combinations thereof. The enzyme additions used to form the plant-based food product may include CPE-1 and CPE-4. CPE-1 protein enzyme including a protease enzyme, which in one example is manufactured by DSM under the product name MaxiPro CPP. CPE-4 is a mixture of a protein enzyme including an acyl transferase made from *Streptomyces mobaraensis* bacteria using microbial fermentation and maltodextrin made from starch derived from tapioca. In one example, the CPE-4 used in the method **100** is manufactured by Ajinomoto Co., Inc. located in Chuo City, Tokyo, Japan. CPE-4, as manufactured by Ajinomoto, is marketed under the product name TI. Alternatively or in addition, other additions may be added to the plant-based fiber product and water, such as fruits-pureed, extracts, or concentrates—and/or a blend of vitamins and minerals, including Vitamins A, C, D, E, K, B1 (thiamine), B2 (riboflavin), B3 (niacin), B5 (pantothenic acid), B6, B7 (biotin), B12 (cyanocobalamin), B9 (folic acid and B9), choline, carnitine and calcium, phosphorus, sodium, potassium, magnesium, manganese, sulfur, chloride, iron, iodine, fluoride, zinc, copper, selenium, chromium, cobalt, and combinations thereof. The plant-based food product may be any of a drink, milk, yogurt, or the like, advantageously having zero sugar content, that is, sugar content between 0 wt % and 0.4 wt %.

[0045] The plant-based food product includes the plant-based fiber product and at least water. Optional additions may be included in the plant-based food product, such as various fruit and food flavorings and spices including strawberry, banana, blueberry, cherry, raspberry, peach, chocolate, vanilla, cinnamon, peppermint, and combinations thereof, plant protein isolates and/or concentrates such as from soy, pea, rice, pumpkin, potato, buckwheat, chia, amaranth, brown rice, fava bean, barley, oat, mushrooms, fungi, and mixtures thereof, enzymes, plant-based flour such as buckwheat flour, brown rice flour, potato flour, barley flour, oat flour, pumpkin flour, chia flour, yellow pea flour, and mixtures thereof, oils such as MCT oil, canola oil, and mixtures thereof, gellan gum, and vitamins such as vitamin A, vitamin D, Vitamin E, Vitamin K, Vitamin C and Vitamins B1, B2, B3, B5, B6, B7, B9, B12, and mixtures thereof. Alternatively or in addition, the plant-based food product may include fruits-pureed, extracts, or concentrates—and/or a blend of vitamins and minerals, including Vitamins A, C, D, E, K, B1 (thiamine), B2 (riboflavin), B3 (niacin), B5 (pantothenic acid), B6, B7 (biotin), B12 (cyanocobalamin), B9 (folic acid and B9), choline, carnitine and calcium, phosphorus, sodium, potassium, magnesium, manganese, sulfur, chloride, iron, iodine, fluoride, zinc, copper, selenium, chromium, cobalt, and combinations thereof.

EXAMPLES

[0046] An example formulation of the base product has the formulation as shown in Table 1. The values shown in Table 1 are meant to include a range of values. For example, each of the components listed in Table 1 may be included in the plant-based food product in a weight percentage of between the value listed in the “Wt % of final product, Low” column and the “Wt % of final product, High” column. This is true regardless of whether the value in the “Wt % of final product, Low” column is greater than or less than the value of the “Wt % of final product, High” column:

TABLE-US-00001 TABLE 1 FORMULATION OF EXAMPLE BASE PRODUCT

Wt % of final product, Low	Wt % of final product, High	Component
69	74	Water
31	26	Oat Flour
0.015	0.035	CSE-10
0.0015	0.0035	CSE-6
0.0015	0.004	CSE-4

[0047] An example formulation of the plant fiber product has the formulation as shown in Table 2. The values shown in Table 2 are meant to include a range of values. For example, each of the components listed in Table 2 may be included in the plant-based food product in a weight percentage of between the value listed in the “Wt % of final product, Low” column and the “Wt % of final product, High” column. This is true regardless of whether the value in the “Wt % of final product, Low” column is greater than or less than the value of the “Wt % of final product, High” column:

TABLE-US-00002 TABLE 2 FORMULATION OF EXAMPLE PLANT FIBER PRODUCT

Wt % of final product, Low	Wt % of final product, High	Component
25	28	Fiber Sludge
90	99	Solids after clarifier
0.1	0.25	CFE-1
0.005	0.01	CSE-6
0.005	0.01	CSE-4
9.89	0.73	Water

[0048] The base product described in Table 1 and the plant fiber product described in Table 2 are combined upon entering the DSI sanitation step **124**, as described above to form a plant-based fiber product. The plant-based fiber product is used in the formulation of the plant-based food product. An example embodiment of the plant-based food product in the form of a plant-based milk has the formulation as shown in Table 3. The values shown in Table 3 are meant to include a range of values. For example, each of the components listed in Table 3 may be included in the plant-based food product in a weight percentage of between the value listed in the “Wt % of final product, Low” column and the “Wt % of final product, High” column. This is true regardless of whether the value in the “Wt % of final product, Low” column is greater than or less than the value of the “Wt % of final product, High” column:

TABLE-US-00003 TABLE 3 FORMULATION OF EXAMPLE PLANT-BASED FOOD PRODUCT

Wt % of final product, Low	Wt % of final product, High	Component
62	64	Water
0.0001	0.0004	CPE-1
0.03	0.0015	CPE-4
0.01	0.005	Various fruit, food flavorings, and spices including strawberry, banana, blueberry, cherry, raspberry, peach, chocolate, vanilla, cinnamon, peppermint
28	31	Plant-based fiber product
6.5	3.5	Soy Protein Isolate
0.15	0.05	Pea Protein Isolate
0.15	0.05	Rice Protein Concentrate
0.15	0.05	Buckwheat Flour
0.15	0.05	Yellow Pea Flour
0.8	0.3	MCT Oil
2	1	Canola Oil
0.09	0.03	Gellan Gum
0.0009	0.0007	Vitamin A
0.00015	0.0001	Vitamin D

[0049] The base product described in Table 1 and the plant fiber product described in Table 2 are combined upon entering the DSI sanitation step **124**, as described above to form a plant-based fiber product. The plant-based fiber product is used in the formulation of the plant-based food product. Another example embodiment of the plant-based food product in the form of a plant-based nutrition shake has the formulation as shown in Table 4. The values shown in Table 4 are meant to include a range of values. For example, each of the components listed in Table 4 may be included in the plant-based food product in a weight percentage of between the value listed in the “Wt % of final product, Low” column and the “Wt % of final product, High” column. This is true regardless of whether the value in the “Wt % of final product, Low” column is greater than or less than the value of the “Wt % of final product, High” column. In Table 4, and throughout this disclosure, “fruit prep” refers to a mixture of the fruit, acidulant, stabilizers, and flavors that are mixed and processed by an ultra-high temperature process.:

TABLE-US-00004 TABLE 4 ALTERNATIVE FORMULATION OF EXAMPLE PLANT-BASED FOOD PRODUCT

	Wt % of final product	Wt % of final Component
Water	15.5	
CPE-1	0.0004	0.00015
CPE-4	0.04	0.015
Various fruit, food flavorings, and spices including strawberry, banana, blueberry, cherry, raspberry, peach, chocolate, vanilla, cinnamon, peppermint		
Plant-based fiber product	32	28
Soy Protein Isolate	8.5	7.5
Pea Protein Isolate	3	2
Rice Protein Concentrate	2.5	1.5
Buckwheat Flour	1	0.5
Yellow Pea Flour	0.5	0.25
MCT Oil	1.2	0.4
Canola Oil	2	1
Gellan Gum	0.08	0.04
Blend of minerals including sodium, potassium, magnesium, manganese, sulfur, chloride, iron, iodine, fluoride, zinc, copper, selenium, chromium, cobalt, and combinations thereof	2.6	2.2
Fruit Prep	31	29

[0050] Notably, all of the example embodiments described above have zero sugar content. To clarify the use of and to hereby provide notice to the public, the phrases “at least one of <A>, , . . . and <N>” or “at least one of <A>, , . . . <N>, or combinations thereof” or “<A>, , . . . and/or <N>” are defined by the Applicant in the broadest sense, superseding any other implied definitions hereinbefore or hereinafter unless expressly asserted by the Applicant to the contrary, to mean one or more elements selected from the group comprising A, B, . . . and N. In other words, the phrases mean any combination of one or more of the elements A, B, . . . or N including any one element alone or the one element in combination with one or more of the other elements which may also include, in combination, additional elements not listed.

[0051] While various embodiments have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible. Accordingly, the embodiments described herein are examples, not the only possible embodiments and implementations.

Claims

1. A method of making a plant-based food product, the method comprising: forming a plant-based fiber product by: adding a plant-based flour and an alpha amylase to water to form a wetted plant-flour mixture; heating the wetted plant-flour mixture to a temperature between 180° F. and 200° F.; cooling the wetted plant-flour mixture to a temperature between 110° F. and 150° F.; adding a glucose oxidase to the wetted plant-flour mixture; clarifying the wetted plant-flour mixture into a plant fiber product and a liquid product; decanting the plant fiber product to form a decanted plant fiber product; sanitizing the decanted plant fiber product by steam injection sanitation to form a sanitized product; and cooling the sanitized product to a temperature between 40° C. and 50° C., resulting in the plant-based food product, wherein the plant-based food product comprises a sugar content of between 0 wt. % and 0.4 wt. %.
2. The method of claim 1, wherein the plant-based flour is oat flour.
3. The method of claim 1, further comprising adding an additional enzyme comprising a protease to the plant-based food product.
4. The method of claim 1, further comprising adding an additional enzyme comprising acyl transferase made from *Streptomyces mobaraensis* bacteria using microbial fermentation to the plant-based food product.
5. The method of claim 1, further comprising adding maltodextrin made from starch derived from tapioca to the plant-based food product.
6. The method of claim 1, wherein sanitizing the decanted plant fiber product by steam injection sanitation to form the sanitized product comprises: feeding the decanted plant fiber product to a direct steam injection feed tank; preheating the decanted plant fiber product to a temperature of between 170° F. and 190° F.; exposing the heated decanted plant fiber product to direct steam injection, further raising a temperature of the decanted plant fiber product to between 280° F. and

290° F., and maintaining at the temperature for at least 4 seconds, to produce the sanitized product.

7. The method of claim 6, further comprising adding fruit to the plant-based food product.
